

Inhom. B.C.:

• ODE: homogeneous

ex:

$$\ddot{y} + \dot{y} + y = 0$$

Non-hom:

$$* \ddot{y} + \dot{y} + y = 5$$

Turn this into a homogeneous ODE:

$$* \hat{y} = y - 5 *$$

$$\hat{y} = y$$

$$\hat{y} = \dot{y}$$

$$\hat{y} + \dot{\hat{y}} + \hat{y} - 5 = 0$$

$$\hat{\ddot{y}} + \hat{\dot{y}} + \hat{y} = 0$$

"Major u ": $u(x,t) = \underbrace{u(x)}_{\text{"shift"}} + v(x,t)$

$$* u_t = u_{xx}, \quad 0 < x < 1, \quad t > 0$$

$$u(0,t) = 0$$

$$u(x,0) = 0$$

Inhom. BC: 

$$* \underline{u(L,t) = 1} *$$

→ "Steady State": is not the same as $\hat{x}_n(x)$
 $v(x,t) = \sum_n \hat{x}_n(x) \hat{T}_n(x)$

$$u(x,t) = u(x) + v(x,t), \quad \text{throw in to PDE } u_t = u_{xx}$$

LHS: $u_t = \frac{\partial u}{\partial t} + v_t(x,t)$

RHS: $u_{xx} = u_{xx} + v_{xx}(x,t)$

$$\bullet \quad v_t(x, \underline{t}) = \underline{u_{xx}} + v_{xx}(x, \underline{t})$$

Similar Sep of var.

$$\bullet \quad v_t(x, t) = v_{xx}(x, t)$$

$$\bullet \quad u_{xx} = 0$$

* Idea: Want u to "absorb" the inhom. B.C.

Apply: BC

$$(i) \quad u(0, t) = 0 = u(0) + v(0, t)$$

$$u(0) = -v(0, t)$$

only way

$$u(0) = 0$$

$$v(0, t) = 0$$

$$* (ii) \quad u(L, t) = \underline{1} *$$

$$u(L, t) = u(L) + v(L, t) = 1$$

$$\Rightarrow u(L) = 1$$

$$v(L, t) = 0$$

I.C.

$$(iii) \quad u(x, 0) = 0 = u(x) + v(x, 0) = 0$$

$$u(x) = -v(x, 0)$$

$$v(x, 0) = -u(x)$$

Solve PDEs: Similar to before

$$(A) \quad u_{xx} = 0$$

$$\leftarrow u(L) = 1, \quad u(0) = 0$$

$$(B)$$

$$v_t(x, t) = v_{xx}(x, t) \leftarrow v(L, t) = 0 = v(0, t)$$

(See notes)

$$v(x, 0) = -u(x)$$

Plug back into

$$u(x, t) = u(x) + v(x, t)$$

Give an outline for inhom.:

$$* \text{ For general } f(x): \text{ In I.C. } u(x, 0) = \underline{f(x)}$$

$$f(x) = u(x) + v(x, 0)$$

$$\rightarrow v(x,0) = f(x) - u(x)$$

Solve as above

Eigen f^n : f^n that satisfies the following eq.

$$Hf = \lambda f$$

Eigen value

Eigen f^n

"Operator"

ex: $\frac{d}{dx}$ is an operator

and $f(x) = e^{2x}$

$$\frac{d}{dx}(f(x)) = \frac{d}{dx}(e^{2x}) = 2e^{2x}$$

ex: $\frac{d^2}{dx^2}$ is an op.

(a) $f(x) = \sin(3x)$

$$\frac{d^2}{dx^2}(\sin(3x)) = -9 \sin(3x)$$

(b) $\frac{d^2}{dx^2}(e^{2x}) = 4e^{2x}$

ex: $u_t - u_{xx} = \underline{e^{-t}}$ $0 < x < 1$, $t > 0$

B.C. $u(0,t) = u(1,t) = 0$

I.C. $u(x,0) = 1$

* Use eigen functions of Herm.

$$X'' = -\lambda^2 X$$

$$\frac{d^2}{dx^2}(X) = -\lambda^2 X$$

$$T' = -\lambda^2 T$$

$$\frac{d}{dx}(T) = -\lambda^2 T$$

$$u_t - u_{xx} = 0$$

$$T(t) X(x)$$

$$\frac{X''(x)}{X(x)} = \frac{T'}{T} = -\lambda^2$$

$$X'' = -\lambda^2 X$$

$$T' = -\lambda^2 T$$

$$v(x,0) = f(x) - u(x)$$

$$X(x) = A \cos(\lambda x) + B \sin(\lambda x)$$

$$X(0) = 0 = A$$

$$X(1) = 0 = B \sin(\lambda)$$

$$\sin(\lambda) = 0$$

$$\Rightarrow \lambda_n = n\pi \quad \leadsto \quad -\lambda^2 = -(n\pi)^2$$

$$X_n(x) = \sin(n\pi x)$$

$$* \underline{u(x,t)} = \sum_{n=0}^{\infty} \underline{b_n(t)} \sin(n\pi x)$$

Before: b_n : a number

Here: $b_n(t) \leftarrow$ we've "absorbed" $T(t)$ from above

* Now, add to the Inhom PDE:

$$* \quad u_t - u_{xx} = e^{-t} *$$

$$u_t = \sum_{n=0}^{\infty} b'_n(t) \sin(n\pi x)$$

$$u_{xx} = \sum_{n=0}^{\infty} b_n(t) (-(n\pi)^2) \sin(n\pi x)$$

$$\leadsto \sum_{n=0}^{\infty} b'_n(t) \sin(n\pi x) + \sum_{n=0}^{\infty} b_n(t) (n\pi)^2 \sin(n\pi x) = e^{-t}$$

$$= \sum_{n=0}^{\infty} [b'_n(t) + b_n(t) (n\pi)^2] \sin(n\pi x) = e^{-t} \cdot 1$$

Want: sum on RHS, with $\sin(n\pi x)$

$$* \quad \boxed{1 = \sum_{n=0}^{\infty} c_n \sin(n\pi x)} * \quad \leftarrow \text{prove that}$$

* using earlier methods: mult. by $\sin(m\pi x)$ then integrate.

$$\sum_{n=0}^{\infty} [b'_n(t) + b_n(t) (n\pi)^2] \sin(n\pi x) = e^{-t} \sum_{n=0}^{\infty} c_n \sin(n\pi x)$$

$$\sum_{n=0}^{\infty} [b'_n(t) + b_n(t) (n\pi)^2] \sin(n\pi x) = \sum_{n=0}^{\infty} e^{-t} c_n \sin(n\pi x)$$

$n=0$

$\cup$

$n=0$

$\cup$

